SAMPLING

Learning Objectives

- Critique different sampling techniques
- Explain the impact a sampling or data gathering method may have on analytical model results
- Recognise methods for estimating summary statistics for a population from a sample

SAMPLING TECHNIQUES

Simple Random Sampling: In this method, each member of the population has an equal and independent chance of being selected for the sample.

Stratified Sampling: The population is divided into distinct subgroups (strata) based on certain characteristics, and then samples are randomly selected from each subgroup proportionally to its size. Similarly, **Cluster Sampling**, where there are a large number of subgroups – a smaller number of clusters are randomly selected first.

Systematic Sampling: A starting point is randomly chosen, and then every "kth" element from the list is included in the sample.

Convenience Sampling: This involves selecting the easiest or most convenient individuals/items to sample.

Purposive (Judgmental) Sampling: Researchers deliberately select specific individuals/items based on their expertise and judgment.

Snowball Sampling: Initially, a few individuals are selected, and then they refer or introduce additional participants. This method is often used in hard-to-reach populations.

ACTIVITY

Critique each sampling technique:

- **Simple Random Sampling**
- **Stratified Sampling**
- **Systematic Sampling**
- **Carrie**
- **Isa**
- **Jason**
- **Kieran**
- **Lewis**
- **Syed**
- **Cluster Sampling**
- **Convenience Sampling**
- **Purposive (Judgmental) Sampling**
- **Garry**
- **Kevin**
- **Mikey**
- **Phil**
- **Rita**

EXAMPLE SOLUTIONS (1)

Critique each sampling technique:

Simple Random Sampling: While simple random sampling ensures each member has an equal chance of being selected, it might not represent certain subgroups well, leading to potential bias in the sample.

Stratified Sampling: While it ensures representation from each subgroup, it can be complex and requires accurate classification of strata, which might not be feasible in some cases.

Systematic Sampling: If there is an underlying pattern in the list, systematic sampling could introduce bias. Also, if the pattern aligns with the sampling interval, it might miss certain population characteristics.

Cluster Sampling: It can lead to increased sampling error due to intra-cluster similarities. Also, if clusters are not homogenous, it might not provide an accurate representation.

EXAMPLE SOLUTIONS (2)

Cont.

Convenience Sampling: Convenience sampling can introduce significant bias since it often excludes hard-to-reach or less-visible portions of the population, leading to results that might not generalize well.

Purposive (Judgmental) Sampling: It can introduce bias due to subjectivity in the selection process, potentially resulting in a skewed representation of the population.

Snowball Sampling: It can lead to overrepresentation of certain groups within the population and might not be suitable for generating a representative sample.

ESTIMATING POPULATION STATISTICS

Let's assume you have a representative sample.

What does the sample tell us about the **population**?

Which of these measures, when calculated from a sample, would give the same result that the population would?

Measures of central tendency:

- Mean
- Mode
- median

Measures of variation:

- Range
- standard deviation
- inter-quartile range

INVESTIGATING HOW FAIRLY SAMPLES CAN REPRESENT A POPULATION

Set up:

We will randomly generate 500 pieces of data from a **normal distribution with a mean of 5 and a standard deviation of 10**.

This will be our **population**!

The mean and standard deviation of the generated data might differ slightly – check these before moving on.

We will then take a sample from it to see if the sample gives the same mean and standard deviation as the population.

Extension: A single sample won't match perfectly – so after we've done one, we'll repeat the process for 100's or 1000's+ of samples to see if, on average, the mean and standard deviation of a sample comes out the same as the population.

VARIANCE

1. Calculate the mean of the data.
2. Find each data point's difference from the mean value.
3. Square each of these values.
4. Add up all of the squared values.
5. Divide this sum of squares by n – 1 (for a sample) or N (for the population).

ACTIVITY: RANDOM NUMBERS

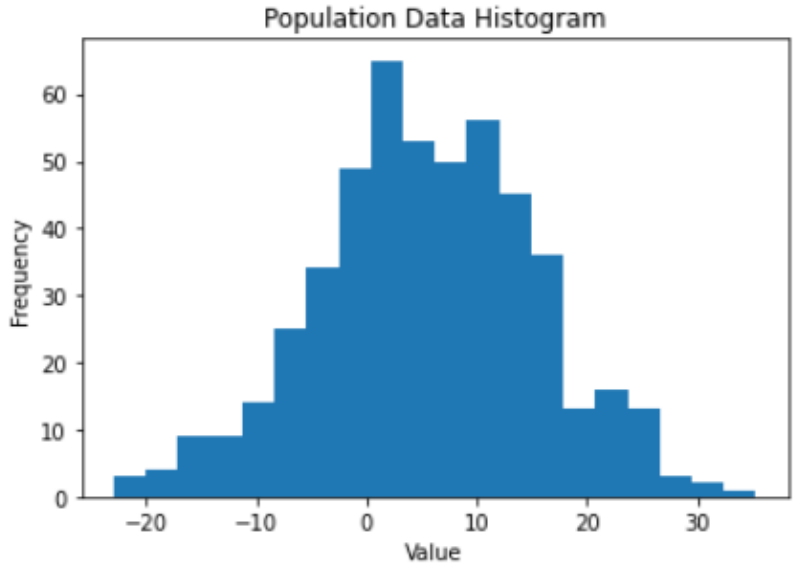
PYTHON

- Import numpy as np and matplotlib.pyplot as plt
- Use np.random.normal() and choose the values of these arguments: ‘loc’ is the mean, ‘scale’ is the standard deviation, ‘size’ is the number of values to generate.
- Create 500 normally distributed random numbers with a mean of 5, a standard deviation of 10. Save this as **populationData**
- Use plt.hist() to visualise the population data.
- Use np.mean() and np.std() to work out the actual mean and standard deviation that was generated.

Note: In real life you may not be able to analyse the whole population – it might mean chopping down every Christmas tree early, opening every bottle of wine in the vineyard, or trying to analyse every social media post from all of your potential customers – not very practical!

RANDOM NUMBERS PYTHON SOLUTION

```python
import numpy as np
import matplotlib.pyplot as plt

MeanToGenerate = 5
SDToGenerate = 10

np.random.seed(1)
populationData = np.random.normal(loc=MeanToGenerate, scale=SDToGenerate, size=500)

plt.hist(populationData, bins=20)  # Adjust the number of bins as needed
plt.title('Population Data Histogram')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.show()

PopMean = np.mean(populationData)
PopSD = np.std(populationData)

print('Population mean:',PopMean)
print('Population sd:',PopSD)
```

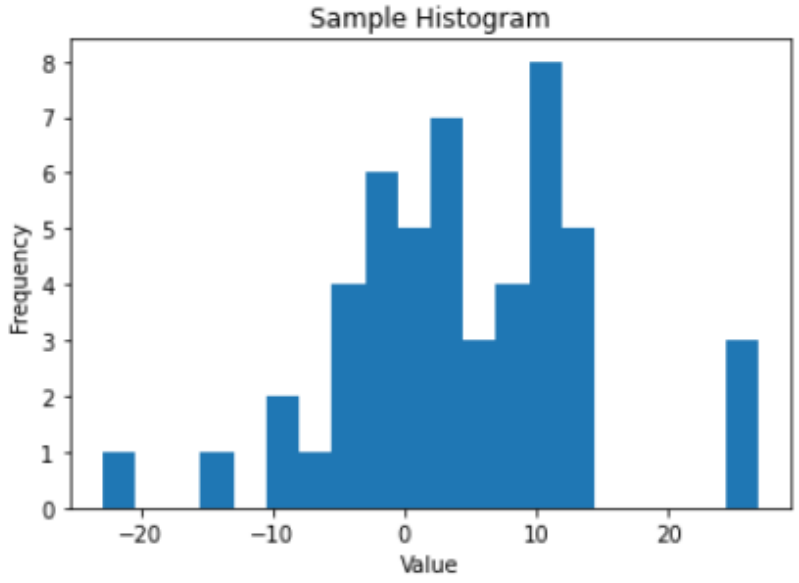



```
Population mean: 5.5343689425630656
Population sd: 9.8840825928265
```

PYTHON SOLUTION

1 SAMPLE

```
np.random.seed(1)
sampleSize = 50
sample = np.random.choice(populationData, size = sampleSize)

plt.hist(sample, bins=20)
plt.title('Sample Histogram')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.show()

SampMean = np.mean(sample)
SampSD1 = np.std(sample)
SampSD2 = np.std(sample, ddof = 1)


print('Sample mean:',SampMean, '. Compared to the population:', PopMean)
print('Sample sd default:',SampSD1, '. Compared to the population:', PopSD)
print('Sample sd ddof=1:',SampSD2, '. Compared to the population:', PopSD)
```




```
Sample mean: 4.357737988094979 . Compared to the population: 5.5343689425630656
Sample sd default: 9.270737308173693 . Compared to the population: 9.8840825928265
Sample sd ddof=1: 9.36485888172677 . Compared to the population: 9.8840825928265
```

PYTHO

EXTEN

REPEA

```python
np.random.seed(1)

numberOfSamples = 100000
sampleSize = 50

sampleMeans = []
sampleSDp = []
sampleSDs = []

for count in range(numberOfSamples):
    sample = np.random.choice(populationData, size = sampleSize)
    sampleMeans.append(np.mean(sample))
    sampleSDp.append(np.std(sample))
    sampleSDs.append(np.std(sample, ddof = 1)) #setting ddof = 1 means the n-1 formula will be used.

print("The sample means average out as", np.mean(sampleMeans), 'compared to the population:', PopMean)
print("The samples standard deviations (n formula) average out as",
      np.mean(sampleSDp),
      'and for the n-1 formula:',
      np.mean(sampleSDs), 'compared to the population:', PopSD)
```

```
The sample means average out as 5.5272770592644465 compared to the population: 5.5343689425630656
The samples standard deviations (n formula) average out as 9.73202902243769 and for the n-1 formula: 9.830833880671399 compared
to the population: 9.8840825928265
```

ESTIMATING THE STANDARD DEVIATION OF THE POPULATION

e.g.
2, 3, 4, 6, 7, 7, 8, 19

Population Variance

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n} = 24.5$$

Population Standard Deviation

$$s = 4.95 \text{ (3 s.f.)}$$

Sample Variance

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = 28$$

Sample Standard Deviation

$$s = 5.29 \text{ (3 s.f.)}$$

While the mean worked great, the sample formula for standard deviation is **tweaked, using *n-1***, to give a better estimate of the variance and standard deviation of the population.

HUMAN OR MACHINE LEARNING FROM SAMPLES

Income	Term	Balance	Debt	Score	Default
17500	Short Tern	1460	272	225	FALSE
18500	Long Term	890	970	187	FALSE
20700	Short Tern	880	884	85	FALSE

The loan_data.csv is a teaching dataset that contains historical data with 856 rows representing loans from a bank. Amongst other features, there is a column that it would be useful to predict in future: Default (True or False) – did they fail to pay off the loan, or did they repay in full.

This data could be treated as a large sample for a bank manager (human) or a machine to learn from – in order that they might consider whether to say yes or no to issuing a loan based on the other columns of data.

What are the potential problems with applying our learning from past experiences or from this dataset in future for new loan customers?

SAMPLING FOR HUMAN OR MACHINE LEARNING - CONSIDERATIONS

- Sample Bias
- Temporal Changes
- Feature Relevance
- Data Quality and Missing Values
- Changing Lending Practices
- Overfitting
- Ethical Concerns
- Concept Drift

HOW DID YOU GET ON?

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